Introduction

1.1 Project aims

The main objective of this doctoral project was to develop the skills required to prepare thin cross sections of fragile crystalline material such that they could be characterised by high resolution scanning transmission electron microscopy (STEM). Subsequently, this technique was applied to a range of novel van der Waals' structures to reveal the nature of buried interfaces. Image processing methods were then developed to extract quantitative information from the STEM images.

1.2 The scope of the thesis

This thesis covers work carried out over a period of 3.5 years of EPSRC and DTRA PhD funding at the School of Materials, Manchester. The literature review chapter discusses the motivation for this work and its place in the rapidly expanding field of 2D materials and their heterostructures. Specifically the chapter will address the challenges associated with focussed ion beam (FIB) sample preparation, STEM imaging of beam sensitive materials and extracting quantitative information from STEM image datasets.

The rest of the thesis focusses on three papers which use cross sectional STEM imaging and bespoke image processing methods to extract quantitative information. The first paper *"Designer capillaries made with atomic scale precision"* relied on cross sectional STEM imaging for quality control whilst Dr Radha Boya developing procedures to create nanoscale capillaries within graphitic stacks. Image analysis of the cross sectional STEM images allowed the pore widths and tortuosity to be characterised.

The second paper *"Measuring interlayer separation in transition metal dischalcogenide heterostructures"* advances the techniques applied in the first paper to characterise complex and beam-sensitive heterostructure systems. Image processing strategies employed alignment and machine learning algorythms to quantitatively analyse the images.

The final paper *"Twinning in van der Waals' materials"* reveals novel bending phenomena in bulk crystals of graphite, hexagonal boron nitride and molybdenum diselenide using cross sectional STEM imaging. Here advanced image processing and analysis techniques allowed me to explore the properties of these bending phenomena and estimate their limits.